

# SAMBP Research Gap Analysis & TR Roadmap

Physics-Informed Protection for IBR-Dominated Microgrids

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**Purpose**    Hiring conversations · Consortium scoping · TR allocation

**Confidential.** This document is an internal working draft intended for hiring and consortium scoping discussions. It contains honest scope boundaries and effort estimates that should not be shared publicly without review.

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# 1. Purpose and Context

This document records the output of a systematic gap analysis of the SAMBP (Substation Adaptive Microgrid Backup Protection) simulation framework, benchmarked against the coverage a Horizon Europe or CIGRE WG B5 consortium would expect from an IBR protection research programme.

It serves three audiences:

- 1. **Hiring:** defines the technical scope for the next research engineer or postdoc — each Tier-1 gap is a self-contained project.
- 2. **Consortium scoping:** identifies which workpackage slots map to which TRs, and which gaps represent genuine collaboration value-adds vs. in-house extensions.
- 3. **Thesis planning:** provides a critical-path view of what must be done before Q3 2028 submission and what can be flagged as future work.

## 1.1. Current Framework in Brief

SAMBP currently delivers four complete relay families, each following a uniform five-layer architecture:

| Relay        | Element                   | Key Result                     | Status |
|--------------|---------------------------|--------------------------------|--------|
| 87L          | Line current differential | 12/12 scenarios, 15 k MC       | Done   |
| 87T          | Transformer differential  | 7/7 events, CT-sat veto        | Done   |
| 87B          | Bus differential          | 6/6 events, $\kappa_n < 6.3$   | Done   |
| OC           | Adaptive overcurrent      | N-gen greedy cascade           | Done   |
| 21/21P       | Mho distance (pilot)      | 5 loc $\times$ 3 types         | Active |
| 40/64G/78/81 | Generator suite           | LOE, sub-harm, GFM out-of-step | Active |

Source types currently covered: synchronous generator (SG), DFIG (Type-3 wind), GFL-PV, GFM inverter. Network topology: radial only (5-node + IEEE 39-bus reference).

# 2. Gap Classification Methodology

Gaps are classified into three tiers:

- **T1 Critical** — a consortium partner will identify this as a missing deliverable; absence weakens the overall thesis claim.
- **T2 Significant** — raised in technical peer review; should be addressed or explicitly bounded as future work with justification.
- **T3 Emerging** — forward-looking; relevant for 2027+ grid standards and positions the lab for follow-on funding.

Effort estimates are in **engineer-months** (EM), calibrated to a postdoc/senior PhD student working on SAMBP code and documentation. One EM assumes  $\sim$ 160 hours of focused research+coding time.

### 3. Full Gap Table

The table below is the primary deliverable of this document. Columns: Gap, Tier, Proposed TR, Effort (EM), Dependencies, and notes on hiring/consortium fit.

| Gap                                                                                                                                                                      | Tier      | TR           | Effort    | Depends on                                                | Hiring / Consortium note                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|-----------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Tier 1 — Critical: Consortium will ask for these explicitly</b>                                                                                                       |           |              |           |                                                           |                                                                                                                                                                |
| <b>BESS source model</b><br>GFM-BESS, VSM mode<br>fault current; near-zero<br>natural fault current;<br>virtual impedance control                                        | <b>T1</b> | <b>TR-68</b> | 3–4<br>EM | <i>TR-05 (87L),<br/>TR-07 (87B); BESS<br/>model lib</i>   | <b>Ideal first hire project.</b><br>Self-contained: write BESS<br>current model, extend 87L/87B,<br>run 1 k MC. Outcome: TR-68 +<br>journal paper.             |
| <b>Type-4 / PMSG<br/>wind</b><br>Full back-to-back<br>converter; negligible<br>neg-seq injection; no<br>crowbar; inrush blocking<br>must be revisited                    | <b>T1</b> | <b>TR-69</b> | 3 EM      | <i>TR-56 (DFIG);<br/>ANDES PMSG<br/>model</i>             | Strong fit for a hire with<br>converter controls background.<br>Pairs with TR-56 ANDES<br>integration already active.                                          |
| <b>Meshed / ring<br/>topology</b><br>Directional relaying (67),<br>pilot schemes, loop-flow;<br>all current models are<br>radial                                         | <b>T1</b> | <b>TR-70</b> | 5–6<br>EM | <i>TR-66 (IEEE<br/>39-bus); network<br/>layer rewrite</i> | Largest structural gap. Requires<br>new network layer in<br>PandaPower; directional OC and<br>pilot logic are new relay families.<br>Consortium WP-level item. |
| <b>Auto-reclose (79) on<br/>IBR feeders</b><br>IEEE 1547-2018<br>ride-through; delayed<br>reclose; live-line /<br>dead-line voltage<br>supervision                       | <b>T1</b> | <b>TR-71</b> | 2–3<br>EM | <i>TR-05 (87L); IEEE<br/>1547 ride-through<br/>study</i>  | Bounded scope; good second hire<br>project. Standalone relay-79<br>FSM. Direct CIGRE WG B5<br>deliverable.                                                     |
| <b>Evolving &amp;<br/>cross-country faults</b><br>SLG → 3-phase<br>escalation; simultaneous<br>faults at two locations;<br>IBR current-limiting<br>complicates detection | <b>T1</b> | <b>TR-72</b> | 2 EM      | <i>TR-05 (87L); fault<br/>scenario engine</i>             | Extends existing fault scenario<br>library. Low implementation<br>risk; high consortium visibility.                                                            |
| <b>Tier 2 — Significant: Should be addressed or explicitly bounded</b>                                                                                                   |           |              |           |                                                           |                                                                                                                                                                |
| <b>Negative-seq<br/>protection under<br/>GFL dominance</b><br>46/47 relay blind spot<br>when GFL is dominant<br>source; neg-seq too small<br>for conventional pickup     | <b>T2</b> | <b>TR-73</b> | 1.5<br>EM | <i>TR-05, TR-07;<br/>GFL dominance<br/>scenario</i>       | In-house extension. Sequence<br>estimator already in <code>sync_oc</code> ;<br>extend to 46/47 threshold study.                                                |

| Gap                                                                                                                                                  | Tier | TR    | Effort   | Depends on                                | Hiring / Consortium note                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|----------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>IEEE 1547-2018 ride-through compliance</b><br>Mandatory ride-through window (momentary cessation vs. continuous); protection must not mal-operate | T2   | TR-74 | 2 EM     | TR-71 (79 relay); GFL/GFM models          | Pairs naturally with TR-71. Consortium standard compliance deliverable.                                             |
| <b>SSCI / Subsynchronous resonance</b><br>DFIG + series-compensated line interaction (5–40 Hz); protection must distinguish resonance from fault     | T2   | TR-75 | 3 EM     | TR-56 (DFIG); series-comp line model      | High academic novelty. ERCOT 2009 case study provides validation baseline. New ANDES model needed.                  |
| <b>COMTRADE / field-data validation</b><br>All validation is synthetic or ANDES-simulated; COMTRADE import pipeline absent                           | T2   | TR-76 | 1.5–2 EM | Data access agreement with utility / EPRI | Single highest-impact credibility move. Requires utility data-sharing MOU. Can be funded as industry collaboration. |
| <b>Open-conductor (series) faults</b><br>Broken conductor, blown fuse; unbalanced supply; dangerous for IBR 3-phase controllers                      | T2   | TR-77 | 1.5 EM   | TR-05 (87L); sequence tools               | Adds 46/open-phase element. Medium effort; strong safety narrative for consortium.                                  |
| <b>Sympathetic tripping — parallel feeders</b><br>Transformer inrush on healthy feeder during fault on adjacent feeder can mal-operate 87T           | T2   | TR-78 | 1 EM     | TR-03 (87T); parallel feeder model        | Short study; extends existing 87T event library with sympathetic-inrush event.                                      |
| <b>N-2 / degraded relay set</b><br>Performance under single-relay failure, CT bypass, or comms loss beyond 87L FSM fallback                          | T2   | TR-79 | 1.5 EM   | TR-05, TR-07; fallback logic              | Directly supports reliability quantification. Extends existing <code>line_fallback_logic.py</code> .                |
| <b>Tier 3 — Emerging: Positions lab for 2027+ funding &amp; standards</b>                                                                            |      |       |          |                                           |                                                                                                                     |
| <b>DC microgrids &amp; hybrid AC/DC</b><br>di/dt discrimination; DC CB coordination; data-centre, EV, marine applications                            | T3   | TR-80 | 5 EM     | New discipline; no current code base      | Scope for a dedicated postdoc hire. Horizon Europe MVDC workpackage candidate.                                      |

| Gap                                                                                                                                              | Tier | TR    | Effort | Depends on                                     | Hiring / Consortium note                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|--------|------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Geomagnetic disturbance (GMD/GIC)</b><br>Quasi-DC GIC causes transformer half-cycle saturation → harmonic currents → 87T mal-op; NERC TPL-007 | T3   | TR-81 | 2.5 EM | TR-03 (87T); GIC transformer model             | High regulatory visibility (NERC TPL-007). Transformer half-cycle saturation model needed in ANDES.   |
| <b>Cyber-physical attacks on GOOSE</b><br>GPS spoofing, GOOSE replay, latency injection on confidence gate; IEC 62351 compliance                 | T3   | TR-82 | 2.5 EM | TR-65 (GOOSE); IEC 62351 framework             | Strong fit for cybersecurity-background hire. Pairs with ch6 security audit code already in codebase. |
| <b>PMU / synchrophasor-based wide-area backup</b><br>C37.118 measurement chain; wide-area backup using PMU data; latency-aware confidence gate   | T3   | TR-83 | 4 EM   | TR-65–67 (system integration); PMU data access | Consortium WP item for grid-scale IBR projects. Requires field PMU data or real-time simulator.       |
| <b>Ferroresonance</b><br>VT / transformer + cable capacitance; sustained oscillation; IBR-islanded systems are susceptible                       | T3   | TR-84 | 2 EM   | TR-03 (87T); non-linear core model             | Extends transformer event library with 8th event. Academic novelty; targeted CIGRE paper.             |
| <b>Cold load pickup &amp; motor starting</b><br>Post-islanding load surge (6–8× rated); protection must not trip; BESS response critical         | T3   | TR-85 | 1.5 EM | TR-68 (BESS); islanding study                  | Small but high operational value. Extends <code>run_islanding_study.py</code> .                       |

## 4. Effort & Priority Summary

### 4.1. Aggregated Effort by Tier

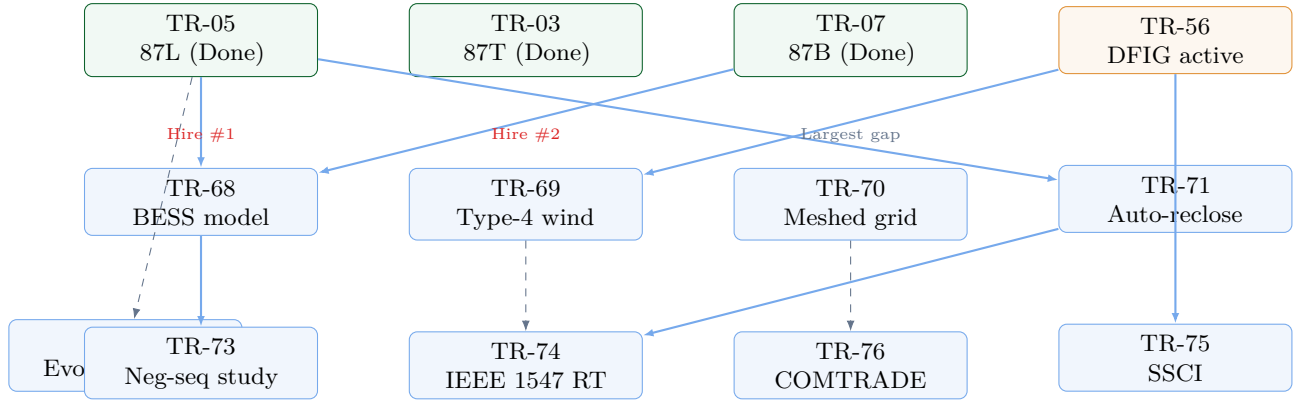
| Tier           | Gaps      | TRs             | Effort range (EM) | FTE-years           |
|----------------|-----------|-----------------|-------------------|---------------------|
| T1 Critical    | 5         | TR-68–72        | 15–18 EM          | 1.5 FTE-years       |
| T2 Significant | 7         | TR-73–79        | 12–15 EM          | 1.2 FTE-years       |
| T3 Emerging    | 5         | TR-80–85        | 17 EM             | 1.4 FTE-years       |
| <b>Total</b>   | <b>17</b> | <b>TR-68–85</b> | <b>44–50 EM</b>   | <b>≈4 FTE-years</b> |

**Thesis scope note.** TR-68 through TR-72 (Tier 1) are candidates for inclusion as Chapter 8 extensions before the Q3 2028 submission deadline, subject to hiring. TR-73 through TR-85 are

scoped as post-thesis consortium deliverables or future-work bounds in Chapter 10 (Conclusions).

## 4.2. Recommended Sequencing

The dependency graph below shows the recommended order of attack. Bold arrows are blocking dependencies; dashed arrows are soft couplings.



## 5. Hiring Profiles

### 5.1. Hire #1 — BESS / Power Electronics Protection (TR-68, TR-74, TR-85)

**Title:** Research Engineer / Postdoc — BESS & IBR Protection

**Duration:** 18–24 months

**Deliverables:** TR-68 (BESS source model + 87L/87B extension), TR-74 (IEEE 1547 ride-through compliance study), TR-85 (cold load pickup). Journal target: *IEEE Transactions on Power Delivery*.

**Required background:**

- Grid-scale BESS control (VSM / droop / virtual impedance modes)
- Python (NumPy/SciPy); familiarity with PandaPower or ANDES
- Protection fundamentals (IEC 60255, percentage-differential relays)

**Nice to have:** Real-time simulator experience (OPAL-RT, RTDS).

### 5.2. Hire #2 — Wind (Type-4) & Subsynchronous Interactions (TR-69, TR-75)

**Title:** Research Engineer — Wind IBR Protection

**Duration:** 12–18 months

**Deliverables:** TR-69 (PMSG Type-4 source model + protection study), TR-75 (SSCI detection and relay discrimination). Journal target: *IEEE Transactions on Energy Conversion*.

**Required background:**

- PMSG / Type-4 converter controls; crowbar-less FRT

- ANDES or PSCAD; understanding of subsynchronous resonance
- Relay sequence-component analysis (Fortescue)

### 5.3. Hire #3 — Network Topology & Wide-Area Protection (TR-70, TR-83)

**Title:** Research Engineer — Meshed Network & Synchrophasor Protection

**Duration:** 24 months (largest scope)

**Deliverables:** TR-70 (meshed topology, directional 67 relay, pilot scheme), TR-83 (PMU / C37.118 wide-area backup). Journal target: *IEEE Transactions on Smart Grid*.

**Required background:**

- Power system network analysis (meshed load flow, fault analysis in loops)
- Directional overcurrent and distance relay coordination
- Synchrophasor (PMU) measurement chains; preferably C37.118

### 5.4. Hire #4 — Cybersecurity & Communications (TR-82, TR-76)

**Title:** Research Engineer — Cyber-Physical Security

**Duration:** 12–18 months

**Deliverables:** TR-82 (GOOSE attack modelling, IEC 62351 compliance), TR-76 (COMTRADE data pipeline, field-data validation MOU). Journal target: *IEEE Transactions on Industrial Informatics*.

**Required background:**

- IEC 61850 GOOSE / MMS protocol internals
- Cyber-physical attack modelling (replay, spoofing, latency injection)
- Familiarity with IEC 62351 security for protection

## 6. Consortium Workpackage Mapping

The table below maps gaps to Horizon Europe / CIGRE workpackage structures. It is intended as a starting point for a consortium proposal — each WP corresponds to one or more hires and a coherent set of TRs.

| WP           | Title                     | TRs                       | Effort       | Partner fit           |
|--------------|---------------------------|---------------------------|--------------|-----------------------|
| WP1          | BESS & GFM Protection     | TR-68, 74, 85             | 6 EM         | Industry (BESS OEM)   |
| WP2          | Type-4 Wind & SSCI        | TR-69, 75                 | 6 EM         | Wind OEM / TSO        |
| WP3          | Meshed Grid & Directional | TR-70, 83                 | 10 EM        | TSO / DSO             |
| WP4          | Standards Compliance      | TR-71, 72, 73, 77, 78, 79 | 10 EM        | Relay vendor          |
| WP5          | Field Validation          | TR-76                     | 2 EM         | Utility (data MOU)    |
| WP6          | Cyber-Physical Security   | TR-82                     | 3 EM         | Cybersecurity partner |
| WP7          | Emerging Topics           | TR-80, 81, 84             | 10 EM        | Academic partners     |
| <b>Total</b> |                           |                           | <b>47 EM</b> |                       |

**Critical path to Q3 2028 submission.** TR-68 (BESS) is the single most important new TR before submission. It closes the largest source-type gap and produces a chapter-length study. Hire #1 must be onboarded by **Q4 2026** to leave sufficient time for TR-68 + journal submission before the thesis lock date.

TR-70 (meshed grid) is the largest structural gap but is *not* on the critical path — it is scoped as post-thesis consortium work and explicitly bounded in Chapter 10.

## 7. Scope Bounds — What Lies Outside the PhD but Inside the Roadmap

### Items excluded from the PhD thesis (SGCRL / IIT Madras scope)

The following are deliberately out of scope for the *thesis*. They are not technical gaps in SAMBP — they are domain boundaries that would widen the thesis beyond its examination scope.

- **Arc-flash detection:** Overlap with occupational safety standards (NFPA 70E / IEC 62271); not a relay protection problem in the SAMBP sense.
- **Stator ground fault (64S) for synchronous machines:** Covered adequately by existing 64G work; marginal additional value within the thesis.
- **Load shedding (UFLS) schemes:** System-stability function, not a selective protection function; separate engineering discipline.
- **PSCAD as primary simulator:** ANDES + PandaPower + OpenDSS cover the required fidelity range for the thesis. PSCAD is used in Ch. 6 data generation only; no additional PSCAD integration is planned within the PhD.

## HVDC Protection — Designated DTaaS Product Line (not excluded; deferred)

**Positioning note (internal only).** HVDC converter protection is *not* declared “not a gap” and is *not* out of scope for SAMBPS DTaaS. It is deferred from the current PhD thesis solely because the thesis examination scope is bounded to AC microgrid and distribution protection. The commercial and collaboration opportunity is real and actively pursued.

HVDC protection represents a natural and high-value extension of the SAMBP Digital Twin-as-a-Service (DTaaS) platform for the following reasons:

- **Technical continuity:** The SAMBP identifiability framework (Stage-1 reduced model + Stage-2 veto) applies directly to VSC-HVDC converter fault models. The IBR current-limiting logic developed in TR-56/TR-62 is a direct precursor to MMC sub-module protection.
- **Collaboration vectors:** The Amprion HVDC corridor and NTU power-electronics protection group represent active engagement opportunities. The DTaaS platform is the natural vehicle for delivering HVDC protection Digital Twin services to these partners.
- **Market differentiation:** HVDC-capable DTaaS is a distinct commercial offering from AC microgrid protection. Branding it as a separate product line (“SAMBPS DTaaS — HVDC Edition”) avoids scope dilution of the PhD while preserving the commercial vector.

**Recommended action.** Initiate a separate HVDC Protection TR stream (TR-H01 onwards, outside the PhD TR-01–TR-100 namespace) under the DTaaS product line umbrella. Target first deliverable: VSC-HVDC fault current model with SAMBP Stage-1 wrapper, benchmarked against CIGRÉ B4 DC grid test system. Potential consortium entry point: Amprion / NTU joint proposal, Q3 2026.

*The PhD thesis (TR-01–TR-100) remains bounded to AC protection. HVDC work proceeds in parallel under the DTaaS product roadmap and does not affect thesis examination scope or timeline.*

## 8. Document Maintenance

This document should be updated:

- When a TR changes status (Planned → Active → Done).
- When a new hire is confirmed — assign TRs and update effort column.
- When a consortium proposal is submitted — attach WP mapping as appendix.
- Quarterly review cadence recommended; owner: Anoop V Eluvathingal.